

a

Three principles

Multi-scale collapsing

raw genotype > fact > claim > view

Evidence chain first-class

claim_id . ruleset@ver . fact_hash

Time as a dimension

commits/ + refs/ = bio diff / bio log

b

.bio bundle layout

```
.bio/
manifest.json
facts/
  <sha256[:2]>/<rest>.json
claims/
  <claim_id>.json
views/
  pgx.md
  germline-clinical.md
commits/<iso-ts>.json
refs/HEAD
```

■ facts: CAS
 ■ claims: interp
 ■ views: LLM-ready
 ■ commits/refs: time

c

Query: "Standard clopidogrel for ACS in CYP2C19 *1/*2?"

format	tokens	pheno	guideline +version	fact hash
VCF	676	—	—	—
.genome	663	●	—	—
Phenopackets	4,001	●	—	—
FHIR R4	13,887	●	●	—
PharmCAT	5,168	●	●	—
.bio	834	●	●	●

.bio is the only format providing all three audit primitives (phenotype, versioned guideline, fact hash) within a sub-1 k token budget.